

Procedure for Evaluation of Blocaïne

(The open source solution for automation)

Table of contents

1 General information.....	2
2 Prerequisites.....	2
2.1 Python3?.....	2
3 Installation of Blocaïne.....	3
3.1 Downloading Blocaïne.....	3
3.2 Installing Python Libraries.....	5
3.2.1 Using ModBus I/O.....	5
3.2.2 Using the OPC Protocol.....	5
4 Network Configuration.....	6
5 Launching the Interpreter.....	7
6 Launching the Block Editor.....	7
7 Verify that Blocaïne is working properly.....	8
7.1 Checking the connection.....	8
7.2 Editing the user block <loop>.....	8
7.3 Compilation, download & execution.....	8
7.4 Debugging.....	8
8 Uninstalling Blocaïne.....	9

1 General information

To evaluate **Blocaïne**, you only need a single PC running two Python interpreters, one for the Blocaïne **Interpreter** and one for the **Block Editor**.

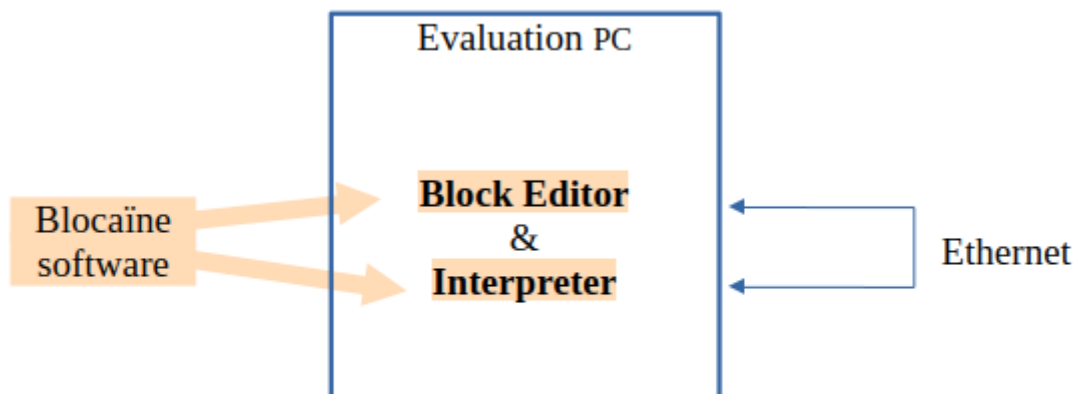


Figure 1 : Evaluation Architecture

2 Prerequisites

You need a computer with at least 100 Mo of free disk and Python3 programming language installed.

2.1 Python3?

Check that Python3 is properly installed on our computer.

On Windows, open a “Command Prompt” (cmd) or a terminal if you are running Linux or macOS, then type the following in the console:

```
python -version
```

ou

```
python3 --version
```

The response should be like :

```
Python 3.xx.xx
```

If it is not the case, install Python 3 on your computer.

3 Installation of Blocaïne

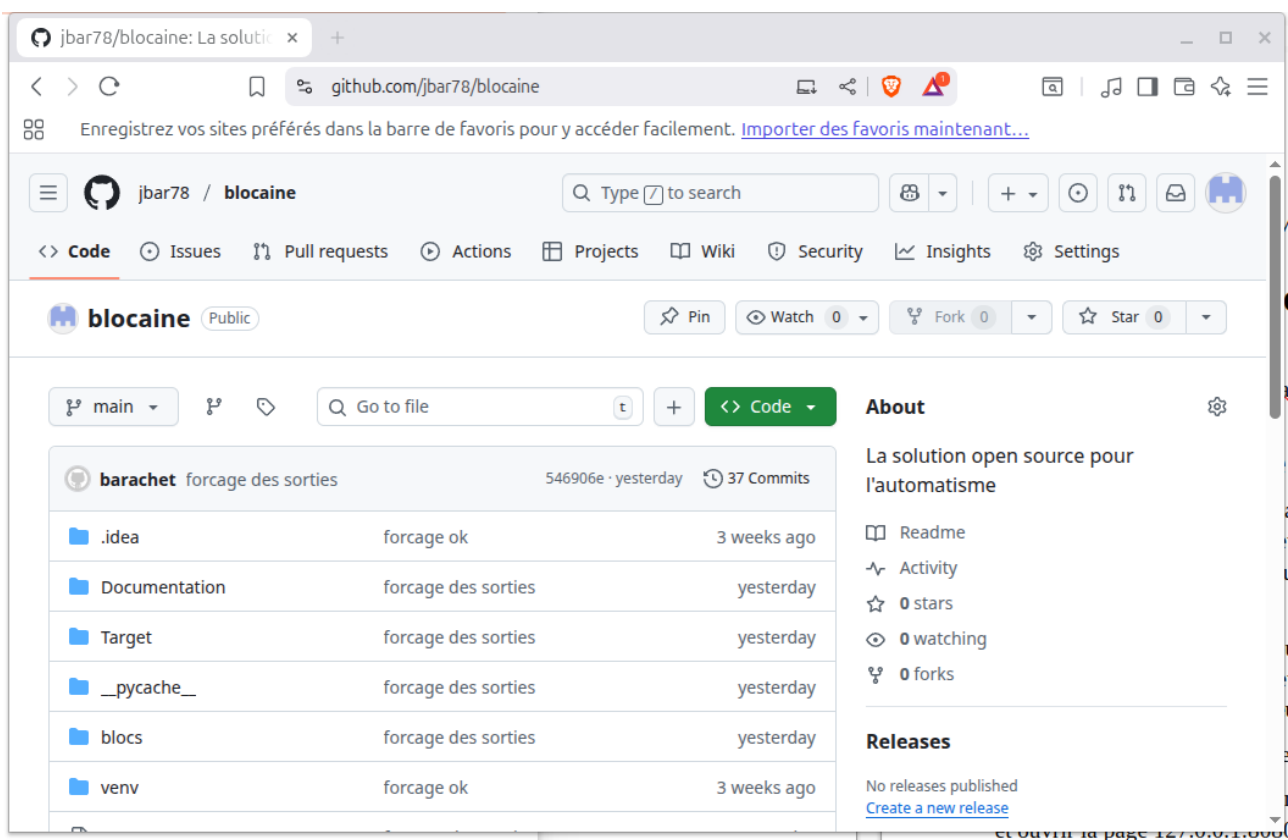
Strictly speaking, there is no real installation process. You simply need to download a ZIP file from GitHub and unzip it. Follow the steps below :

3.1 Downloading Blocaïne

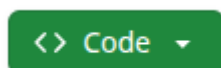
Using your browser, go to:

<https://github.com/jbar78/blocaïne.git>

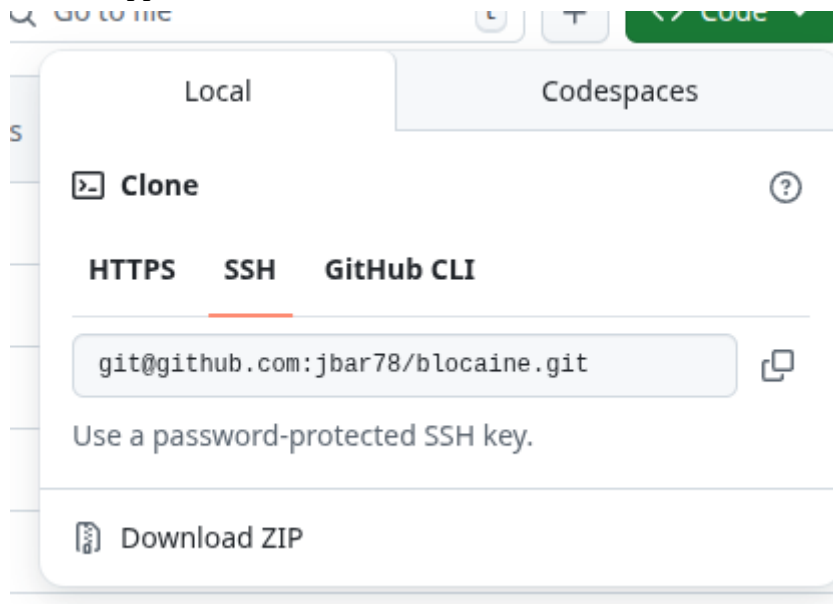
You will arrive on the following page :



Click the main green « code » button.



A popup window will appear. :



Click « Download ZIP », select here you want to save ZIP file and once downloaded, unzip it. you should then have a new directory named «blocaine-main ».

3.2 Installing Python Libraries

The version of Blocaïne you have just downloaded does not require any library (module) other than the standard libraries installed with Python 3, because the ModBus and OPC features have been disabled to make installation easier.

Blocaïne is therefore ready to use.

3.2.1 Using ModBus I/O

If you want to use “ModBus” hard I/O, you must enable this feature by editing the file `PARAM_CONFIG.py` located in the `Target` directory, changing the value of the `PARAM_CONFIG_MODULE_MODBUS` parameter from `False` to `True`, then copy this file into the root directory `blocaïne-main`. You must also install the following library (module): `pymodbus` as follows:

On Windows, type in the console:

```
pip install pymodbus
```

On Ubuntu, type in the console:

```
sudo apt install python3-pymodbus
```

On other Linux distributions, type:

```
python3 -m pip install pymodbus
```

3.2.2 Using the OPC Protocol

If you want the Blocaïne executor to include an OPC UA server so that it can connect to your HMI, you must enable this feature by editing the file `PARAM_CONFIG.py` located in the `Target` directory, changing the value of the `PARAM_CONFIG_MODULE OPCUA` parameter from `False` to `True`, then copy this file into the root directory `blocaïne-main`. You must also install the following library (module): `opcua` as follows:

On Windows, type in the console:

```
pip install OPCUA
```

On Ubuntu, type in the console:

```
sudo apt install python3-OPCUA
```

On other Linux distributions, type:

```
python3 -m pip install opcua
```

4 Network Configuration

You don't need to do anything, since the Target's IP address of HTTP and TCP/IP servers of **interpreter** are by default set to: 127.0.0.1

Note: the target address is defined by the parameter `PARAM_TCP_TARGET_IP = "127.0.0.1"` contained in the `PARAM_NETWORK.py` files located in the root of the `blocaïne-main` project directory and in the `Target` directory; these two files must be identical.

5 Launching the **Interpreter**

- On Linux: Open the folder “blocaine-main/Target” in the file explorer, right-click on “launcher.sh”, then select “Run as a program.”
The Python3 interpreter will start, and a terminal window will appear.
- On Windows: Open the folder “blocaine-main/Target”, double-click “main.py.” If the firewall detects new connections, click Allow.
- On Mac: Run “main.py” inside the “blocaine-main/Target” directory using the Python3 interpreter.

Once launched, the **Interpreter** is operational. You can now open your web browser and go to <http://127.0.0.1:8080> (port 8080) to navigate through the **Interpreter**’s menu

6 Launching the **Block Editor**

- On Linux: Open the “blocaine-main” directory in your file explorer, right-click on “launcher.sh,” then select “Run as a program.”
- On Windows: Open “blocaine-main”, then double-click “main.py.”
- On Mac: Run “main.py” in the “blocaine-main” project directory using Python3

The Python interpreter will start, and both a terminal and the Block Editor’s graphical interface will appear. The **Block Editor** automatically connects to the Blocaïne **interpreter**.

You can now create your first **user** blocks, upload them, and execute them in the **interpreter**.

7 Verify that Blocaïne is working properly

7.1 Checking the connection

To check whether the Executor is properly connected to the Block Editor, look at the status bar located at the bottom of the Block Editor's graphical interface. It should display the message "Target: Connected."

7.2 Editing the user block <loop>

Using the Block Editor, edit the user block <loop> via the "Block" menu. Then click "Open," select the file "loop.bloc," and click "Open." The block should appear in the graphical area.

7.3 Compilation, download & execution

To compile the user block <loop>, download it to the Executor, and run it, use the "Target" menu. Then click "Check, Transfer & Execute <loop>" or click the ⚡ icon.

After compilation, a popup will ask you to confirm whether you want to download and run the <loop> block. Click "Yes."

You can verify that the <loop> block has started correctly by using the web server (<http://127.0.0.1:8080>).

7.4 Debugging

To view the current values of the variables in the <loop> block, enable "Monitoring" mode via the "Target" menu. Then click "Monitoring (Start/Stop) [Space]," press the Space key, or click the 👁 icon.

8 Uninstalling **Blocaïne**

Since installation only involves downloading and unzipping a ZIP file to obtain the “blocaïne-main” directory, uninstalling Blocaïne simply consists of deleting the ZIP file and the “blocaïne-main” folder.